

RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Information Technology
First Year - Semester I Examination – February 2024

ICT 1404 – MATHEMATICS AND STATISTICS FOR COMPUTING

Time: Three (03) hours

Instructions

- Answer any **FIVE (05)** questions.
- **Calculators are allowed.**
- Each question carries equal marks.
- Statistical tables and formulas are provided as attachments.

- 1) a) i) Find the number of words with or without meaning which can be made using all the letters of the word COLOMBO.
ii) If these words are written as in a dictionary, after how many words the first word begins with letter 'O' comes?
iii) What is the 40th word?
(10 Marks)
- b) In a software firm, there are 50 employees, of which 22 employees have at least 5 years of experience. In a staff development programme, 22 employees are to be chosen for a foreign training, of which at least 20 employees must have at least 5 years of experience. In how many ways the choice can be made?
(10 Marks)
- 2) a) If $A = \begin{pmatrix} 5 & 2 \\ 5 & 4 \end{pmatrix}$ and $A = \begin{pmatrix} 1 & -2 \\ 3 & 5 \end{pmatrix}$.
Find (i) $A+B$ (ii) AB (iii) A^{-1}
(06 Marks)
- b) Using matrices or Cramer's rule solve the following linear equations.
 $x+3y+z=15$, $3x+4y+3z=30$, $9x+y+7z=49$
(14 Marks)
- 3) a) Calculate the followings, given $P(A) = 0.35$ and $P(B) = 0.65$:
i) $P(A \cap B)$ when A and B are independent.
ii) $P(A \cup B)$ when A and B are not mutually exclusive.
(08 Marks)

- b) An automobile manufacturer has three factories: A, B, and C. They produce 50%, 30%, and 20% respectively, of a specific model of car. The percentages of white colour cars produced in factories A, B and C are respectively 30%, 40% and 25%.
- If an automobile produced by the company is selected at random, find the probability that it is white.
 - Given that an automobile selected at random is white, find the probability that it came from factory B. (12 Marks)
- 4). a) It has found 80% of undergraduates use digital ID cards. Find the probability that in a randomly selected sample of five undergraduates,
- Exactly two undergraduates use digital ID cards,
 - More than two undergraduates use digital ID cards (10 Marks)
- b) On average two accidents were reported in a certain part of highway road during a month. Assuming that the number of accidents can be modeled by a Poisson distribution, find the probability that it happens
- Exactly 2 accidents in a given month,
 - At least two accidents in a given two-week period. ($e^{-2}=0.135$) (10 Marks)
- 5) The weight of a loaf of bread which is normally distributed with a mean 450g and standard deviation 10g. Find the probability that the weight of a randomly selected loaf of bread is
- less than 436.5g
 - less than 440g.
 - between 435g and 455g
 - more than 465g (20 Marks)
- 6) The information of crop yields and amount of average rainfall in 7 years duration is given by the following table.

Average rainfall, x (cm)	Crop yields, y. (tons per acre)
12	60
14	80
15	80
13	50
14	70
12	90
11	60

- Calculate the equation of line of regression of y on x using least square method. (10 Marks)
- Estimate the amount of the crop when the Average rainfall is 10cm. (04 Marks)
- Compute the coefficient of Determination. (06 Marks)

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Statistical Formulae

Probability

If A, B are mutually exclusive events $P(A \cup B) = P(A) + P(B)$

If A, B are not mutually exclusive events $P(A \cup B) = P(A) + P(B) - (A \cap B)$

If A, B are independent events $P(A \cap B) = P(A)P(B)$

Bayes' Theorem

$$P(A|L) = \frac{P(L|A)P(A)}{P(L|A)P(A) + P(L|B)P(B) + P(L|C)P(C)}$$

Probability Distributions

Normal Distribution $Z = \frac{X - \mu}{\sigma}$

Binomial Distribution $P(x) = {}^nC_x p^x q^{n-x}; \quad q=1-p, x=0,1,2,\dots,n$

Poisson Distribution $P(x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad \lambda = \text{mean}$

Regression Analysis

Line of Regression $y = a + bx \quad b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} \quad a = \bar{y} - b\bar{x}$

Coefficient of Determination, $R^2 = b^2 \left(\frac{\sum x^2 - n\bar{x}^2}{\sum y^2 - n\bar{y}^2} \right), \quad R^2 = b^2 \left(\frac{n \sum x^2 - (\sum x)^2}{n \sum y^2 - (\sum y)^2} \right)$

Standard Normal Table for negative Z values

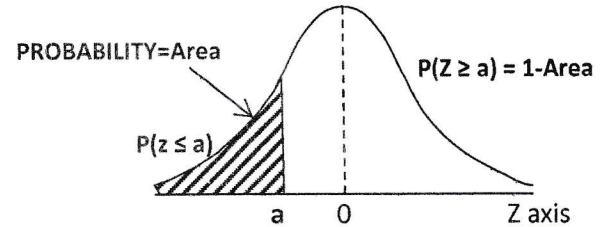


Table value = Area

TABLE A										
Standard normal probabilities										
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Standard Normal Table for positive Z values

PROBABILITY=Area

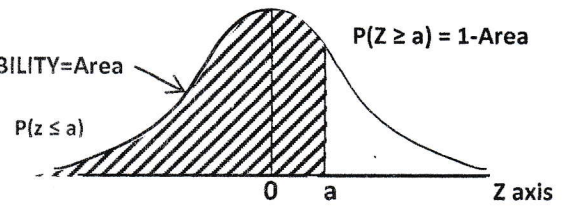


Table value =Area

TABLE B Standard normal probabilities (continued)										
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998